Mehrshad Bagherebadian

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Education

Michigan State University East Lansing, MI
B.S. in Computer Science, Concentration in Artificial Intelligence *Expected Dec. 2026*
Minors: Data Science and CMSE; Business Cognate

Coursework: Machine Learning, Artificial Intelligence, AI Agents, Big Data Analysis, Data Science Methods, Applied ML, Software Engineering, Web Development & Architecture, Probability & Statistics

Experience

Software Developer — The Capstone Experience Aug. 2026 – Present
Michigan State University — Urban Science *East Lansing, MI*

- Developing an AI-powered dealership analytics platform with Angular, TypeScript, C#/.NET, SQL Server, and Python/FastAPI; applying machine learning and AI agents to relate offer characteristics, customer distance, and service history to response and redemption rates, informing campaign targeting, pricing, and timing
- Building a multi-coupon extraction pipeline with Azure Document Intelligence OCR, Microsoft AI Foundry, LangChain, and LangGraph, combining structured LLM extraction, field-level validation, and human review
- Developing interactive dashboards and geographic market views to benchmark campaign effectiveness against competitors and in-network dealerships, with Docker deployment to an Azure VM through GitLab CI/CD

Undergraduate Researcher Aug. 2026 – Present
Michigan State University — Advisor: Dr. Mohammad M. Ghassemi *East Lansing, MI*

- Developing a biomedical NLP question-answering agent with evidence retrieval and confidence-aware selective answering to detect unsupported responses and defer uncertain cases for human review
- Evaluating accuracy, answer coverage, and hallucinations through controlled experiments with complete, incomplete, and noisy evidence; preparing research for EACL 2027 Student Research Workshop and MSU UURAF

IT Software Developer Intern Dec 2025 – Aug 2026
eMortgage Funding LLC *Troy, MI*

- Automated mortgage data processing and loan validation in Python, cutting manual processing time by about 30%
- Implemented and maintained Salesforce CRM workflows, troubleshooting data synchronization issues and improving data consistency across 500+ loan records
- Built SQL queries for data validation and backend troubleshooting, debugging REST API/JSON integrations and conducting functional and regression testing to ensure data accuracy

Projects

Perturvia – An Open-Source AI Agent Reliability Engineering Platform Oct. 2026 – Present
Python, FastAPI, Next.js, TypeScript, PostgreSQL, MCP, LangChain, LangGraph, scikit-learn, AWS

- Building a full-stack platform to test, monitor, and improve AI agent reliability across LLM workflows, external tool integrations, and multi-agent systems
- Designing a Model Context Protocol (MCP) gateway for distributed tracing, observability, fault injection, and execution replay to reproduce failures, identify root causes, and evaluate agent resilience
- Developing RESTful APIs, PostgreSQL models, and a Next.js dashboard for agent management and reliability analytics, with planned ML-driven failure prediction, counterfactual analysis, and adaptive recovery policies

AI-Powered Claims Timeline Generator | MSU AI Club | Auto-Owners Insurance Aug. 2026 – Present

- Developing an AI-powered insurance claims platform using Python, FastAPI, Streamlit, and Azure Document Intelligence OCR to extract events from unstructured documents and generate chronological, source-cited timelines
- Building a Retrieval-Augmented Generation (RAG) pipeline with LlamaIndex, ChromaDB/Chroma Cloud, OpenAI embeddings, and OpenRouter LLMs for semantic search, evidence verification, structured extraction, event deduplication, and date resolution

Technical Skills

Languages: Python, TypeScript, JavaScript, C#, C++, SQL, HTML/CSS
Frameworks: FastAPI, React, Next.js, Tailwind CSS, Angular, Flask, ASP.NET Core, SQLAlchemy, Pydantic, Streamlit
AI & ML: PyTorch, TensorFlow, scikit-learn, LangChain, LangGraph, LlamaIndex, RAG, MCP, pandas, NumPy
Databases: PostgreSQL, SQL Server, MySQL, ChromaDB/Chroma Cloud
Cloud & DevOps: Azure, AWS, GCP, Docker, Git, GitHub Actions, GitLab CI/CD, Prometheus